Tier 1 — Channel Validation: Plan of Attack

Validate every fibre model against published data — make the simulator defensible in peer review

Task 1 — CD Validation [Agrawal Fig 2.6]

- Launch Gaussian pulse through MZM (sigma=5-30 ps)
- Pass through cable() at z/L_D = 0.5, 1.0, 2.0
- Fit output |E|^2 to Gaussian, measure sigma(z)
- Compare to analytic: sigma(z) = sigma_0 * sqrt(1 + (z/L_D)^2)
- Target: width ratio error < 0.1%
- Output: val_cd_{seed}.png, val_cd_{seed}.csv

Task 2 — PMD Validation [Razavi Fig 2.11]

- Current state: fiber.py L145 uses np.random.rayleigh (2D magnitude)
- Generate 10^4 realisations with PMD only
- Histogram DGD vs BOTH Rayleigh and Maxwell PDFs (let data decide)
- Compute <DGD>, confirm <DGD> = PMD_coeff * sqrt(L)
- If Rayleigh != literature expectation -> fix to Maxwell
- Output: val_pmd_dgd_{seed}.png, val_pmd_{seed}.csv

Task 3 — Attenuation Validation [Keiser Eq 3.6, SMF-28]

- Sweep L = 0-200 km, measure P_out / P_in
- Overlay ideal: 10^{-alpha*L/10} at 0.182 dB/km
- Output: val_att_{seed}.png, val_att_{seed}.csv

Task 4 — Birefringence Validation [Yuan Fig 1]

- Sweep bend radius R = 5-100 mm at T = 25 C
- Plot beat length L_B = lambda / Delta_n vs R
- Delta_n_0 = {0.5, 1.0, 2.0} x 10^{-5}
- Confirm L_B propto R / Delta_n_0 scaling
- Output: val_biref_{seed}.png, val_biref_{seed}.csv

Task 5 — Reproducibility Pass

- All scripts save PNG + CSV with {name}--seed{seed}--{params}.{ext}
- New run_all.py executes all 4 validations with --seed 42

Estimated Effort

Task	Days	File
1. CD validation	1	validate_cd.py
2. PMD validation (+ fix)	1.5	validate_pmd.py + fiber.py
3. Attenuation validation	0.5	validate_attenuation.py
4. Birefringence validation	1	validate_birefringence.py
5. Reproducibility pass	1	run_all.py, filename conventions
Total	5 d	4 validated models + run_all.py

Key Decisions

- PMD: Validate current Rayleigh sampling first, compare to Maxwellian literature - fix only if needed
- No cable() API changes expected (unless PMD fix changes DGD sampling)
- All scripts accept --seed for reproducibility
- If fiber.py is changed, create a git checkpoint first